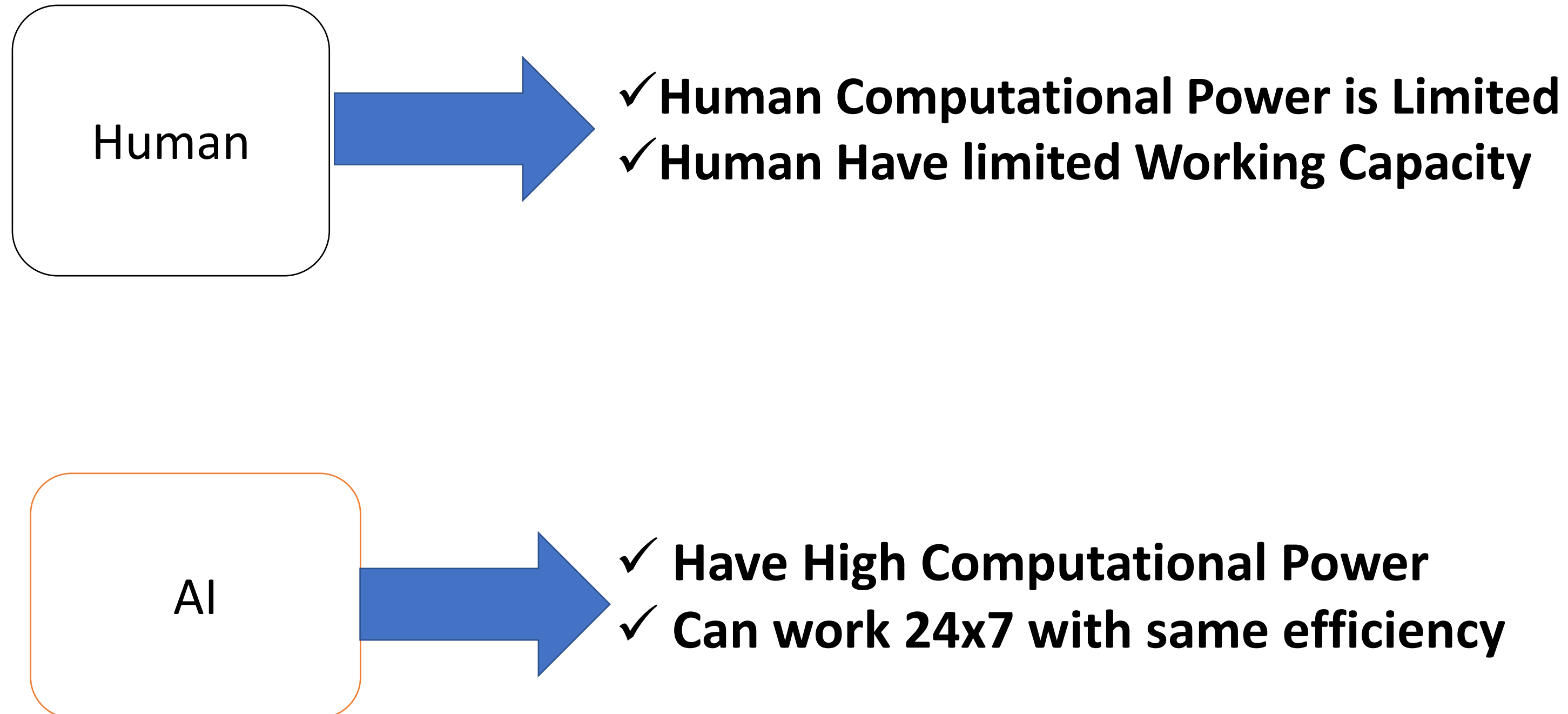


Machine Learning Basics

Chapter :1

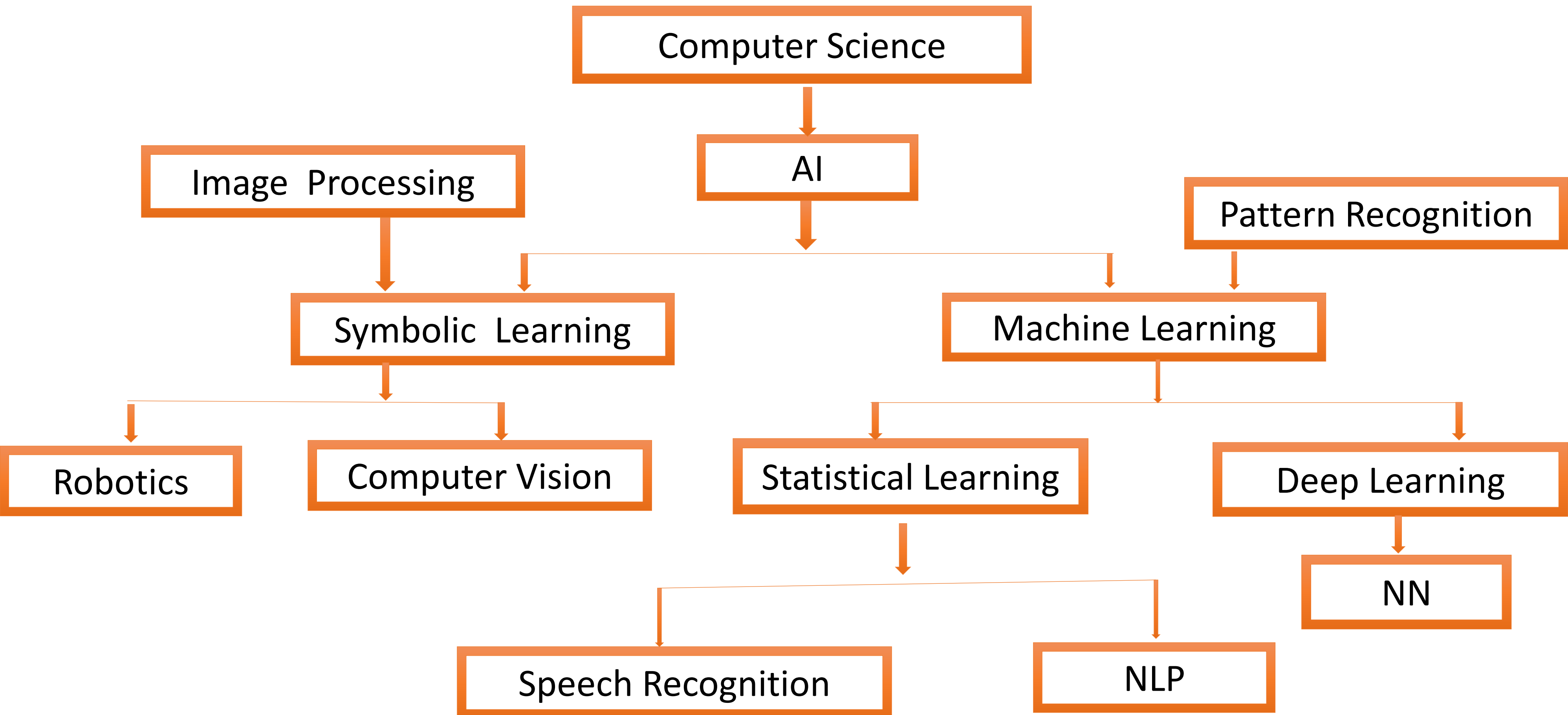
Why Artificial Intelligence (AI)?



What is Artificial Intelligence (AI)?

- Artificial intelligence (AI) is a term to describe a branch of computer science that is dedicated to creating intelligent machines that would learn to work and react like humans.
- The birth of artificial intelligence 1952–1956
- In the 1940s and 50s – AI worked on mathematics, psychology, engineering, economics and political science.

- In 1950 Alan Turing-: **Turing's test** it was reasonable to say that the machine was "thinking"
- In 1951 **Game AI**
- In 1955 **Symbolic reasoning**
- The golden years 1956–1974:- The programs developed in the years(NLP)
- In 1967 **Robotics**
- Boom 1980–1987 AI program called "expert systems"
- **Artificial neural network (ANN)**
- **Machine Learning**
- **Deep Learning** and Research continue on AI



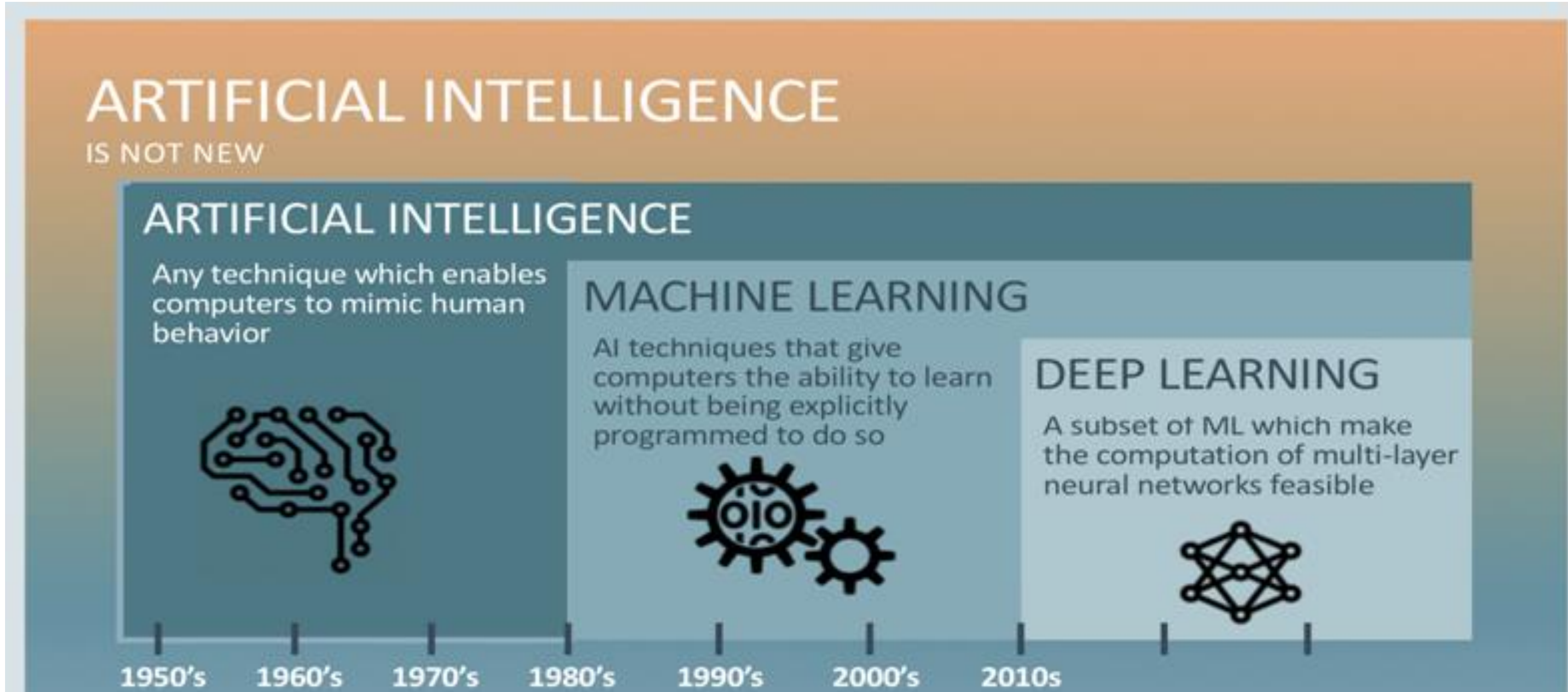
Machine Learning based on Databases, Symbol and Computer Vision

- Machine Learning based on Databases. Data are required to train machine and analysis of data using Machine Learning Algorithm.
- Symbolic Learning based on symbol like human readable character, number and other symbol.
- **Computer vision** tasks include methods for acquiring, processing, analyzing and understanding digital images, and extraction of high-dimensional data from the real world in order to produce numerical or symbolic information, e.g. in the forms of decisions.

Application of AI



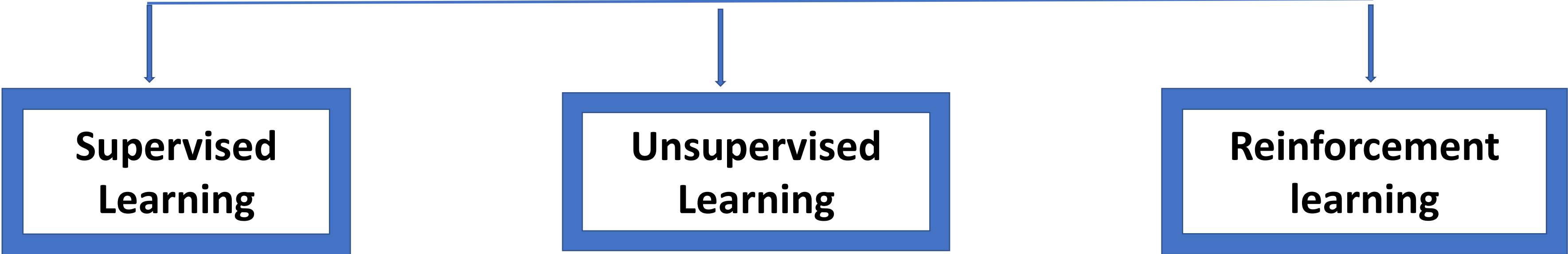
Machine Learning



Machine Learning:

- It is Subset of Artificial Intelligence
- It is brain of Artificial Intelligence
- It is collection of Algorithm
- It provides systems the ability to automatically learn and improve from experience without being explicitly programmed.
- Machine learning focuses on the development of computer programs that can access data and use it learn for themselves.

Type of Learning



Supervised Learning

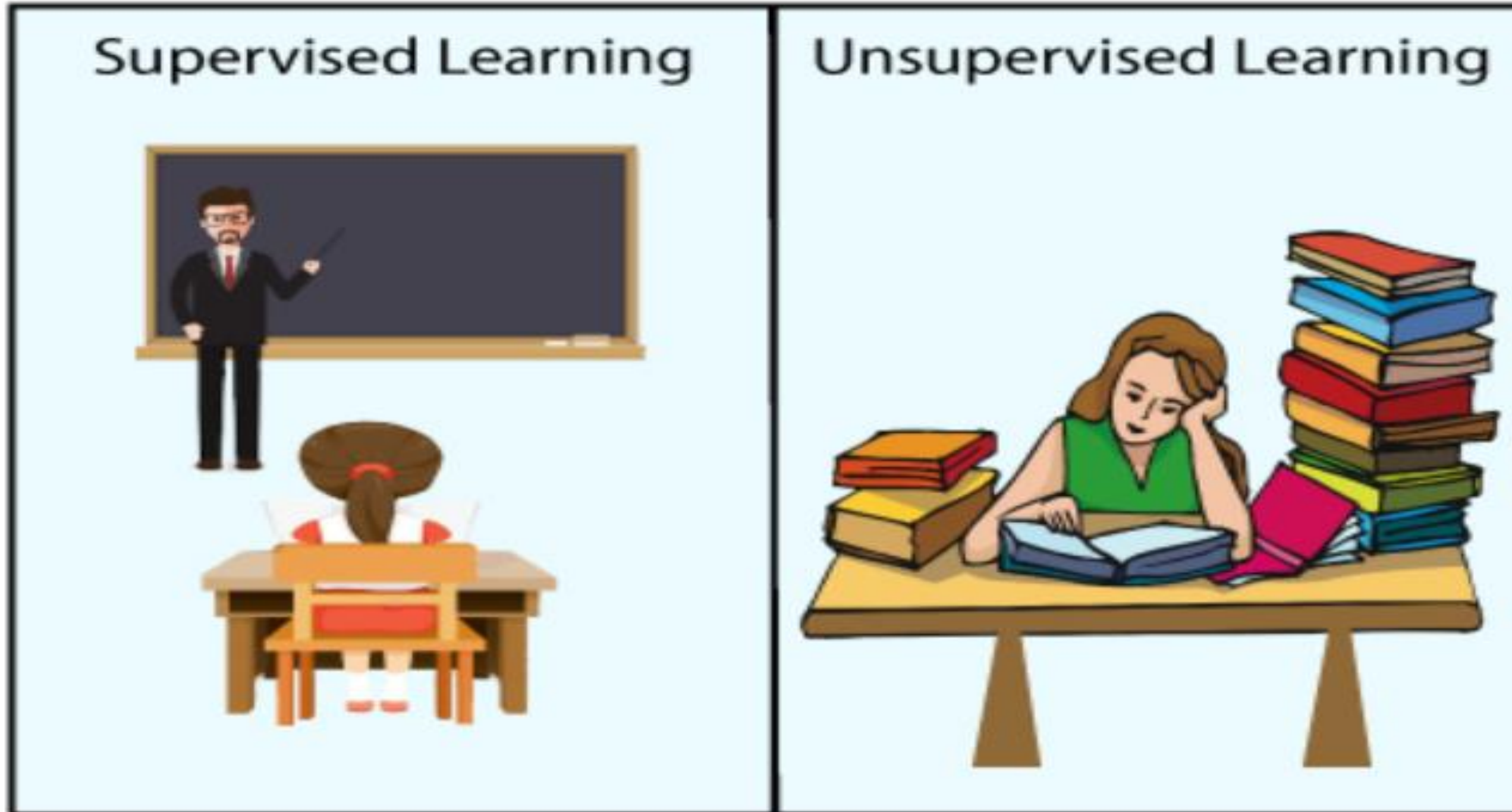
Supervised learning is the machine learning task of learning a function that maps an input to an output based on example input-output pairs.

Unsupervised Learning

Unsupervised learning is a type of **machine learning** algorithm used to draw inferences from datasets consisting of input data without labeled responses.

Reinforcement learning

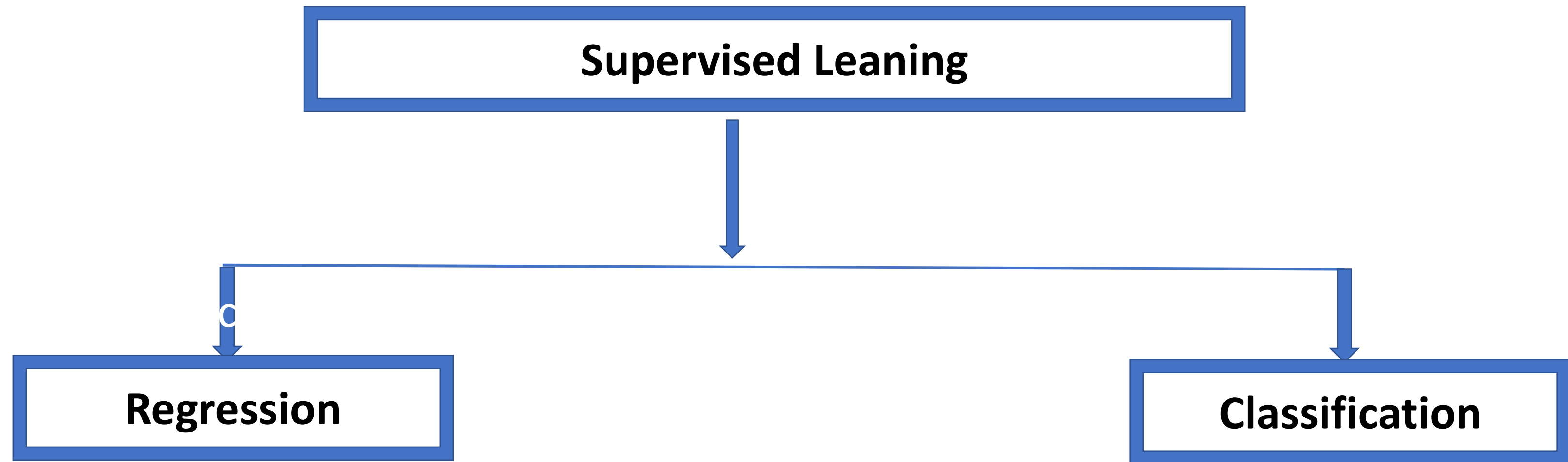
Reinforcement learning (RL) is an area of **machine learning** concerned with how software agents ought to take actions in an environment in order to maximize the notion of cumulative reward.



Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.

Supervised Learning	Unsupervised Learning
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.
Supervised learning needs supervision to train the model.	Unsupervised Learning can be classified in Clustering and Associations problems.
Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.

Supervised Learning	Unsupervised Learning
Supervised learning model produces an accurate result.	Unsupervised learning model may give less accurate result as compared to supervised learning.
Supervised learning is not close to true Artificial intelligence as in this, we first train the model for each data, and then only it can predict the correct output.	Unsupervised learning is closer to the true Artificial Intelligence as it learns similarly as a child learns daily routine things by his experiences.
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorithm.



- Numerical Value Analysis
- Example Predicting age of a person
- Predicting whether stock price of a company will increase tomorrow

- Categorical or discrete Data Analysis
- E-Mail Data "spam" or "not spam."

Machine Learning Algorithm

- Linear Regression.
- Logistic Regression.
- Decision Tree.
- SVM.
- Naive Bayes.
- kNN.
- K-Means.
- Random Forest.

Thank You!

